Case Reports

Pasteurella multocida septicemia during pregnancy

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Documented cases of septicemia due to *Pasteurella multocida* during pregnancy are rare. Only three cases have previously been reported in the English-language literature;[1-3] all occurred in the third trimester. None of the women died, but in two cases the fetus died, apparently as a result of the infection.

Case report

A 22-year-old gravida 2, para 1 woman was seen in the emergency department of a community hospital at 37 weeks' gestation. She had experienced a normal antenatal course and 5 years previously had given birth to a live girl. She had taken no medications during pregnancy and had no history of alcohol or drug abuse. An hour and a half earlier she had suffered a bite on her left thumb while separating two cats that were fighting. There was a single red spot on the thumb. Because her immunization was unclear, tetanus toxoid was administered. The wound was irrigated with chlorhexidine gluconate and cetrimide, and she was discharged.

The next evening the woman presented with crampy lower abdominal pain. Her heart rate was 128 beats/min, respiratory rate 28/min, temperature 36.6°C and blood pressure 104/40 mm Hg. A systolic murmur was heard, but there was no evidence of cardiomegaly. Her throat and chest were clear. The gravid uterine distention was consistent with the length of gestation. The fetal heart rate was regular, at 144 beats/min. Despite abdominal tightening, the cervix was not dilated, and no bleeding or rupture of the fetal membranes was evident.

Laboratory tests gave the following values: hemoglobin concentration 145 g/L, erythrocyte count 4.6×10^{12}/L, hematocrit 0.43%, mean corpuscular volume 94 fl and leukocyte count 4.9×10^9/L. Wright-stained peripheral blood films revealed toxic granulation and vacuolation of neutrophils as well as intracellular and extracellular coccobacilli (Fig. 1).

Occasional irregularly contracted erythrocytes were present, with evidence of fragmentation. There were 5 nucleated erythrocytes per 100 leukocytes. The platelet count was 34×10^9/L. Urinalysis showed a pH of 5, occasional erythrocytes and 10 to 15 leukocytes per high power field.

Two hours after admission the patient had a single episode of vomiting. Her temperature was 38.5°C. Antipyretics and antiemetics were given,

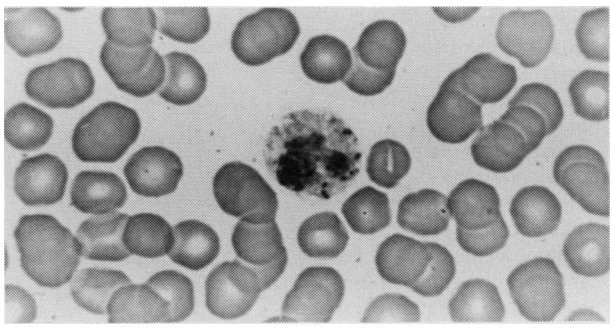


Fig. 1 — Photomicrograph of blood film, showing polymorphonuclear leukocyte with intracellular bacteria and toxic vacuolation (Wright's stain; ×1200, reduced approximately 30%).

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but she vomited again and had diarrhea. Intravenous fluids were administered. Her blood pressure remained stable, and the fetal heart rate remained regular, at 154 beats/min. Neither abdominal contraction nor uterine tenderness was evident. It was thought that she had viral gastroenteritis, which was locally prevalent, and was not in labour.

Four and a half hours later the patient's temperature was 39.3°C and her pulse rate 115 beats/min. Two blood specimens were drawn for culture, and ampicillin, 1 g given intravenously every 6 hours, was prescribed. Arrangements were begun to transfer her to Sault Ste. Marie, Ont. Two and a half hours later her temperature was 38°C, pulse rate 120 beats/min and blood pressure 120/70 mm Hg. The fetal heart rate was 160 beats/min. Half an hour later she again complained of crampy lower abdominal and right lower quadrant pain. Within the hour, before she could be transferred, she experienced difficulty breathing, vomited a small amount of yellow fluid, lost consciousness and died. The fetus did not survive.

Postmortem examination of the woman showed features of gram-negative septicemic shock with disseminated intravascular coagulation, bilateral adrenal hemorrhage and proximal tubular necrosis. Histopathologic examination revealed microthrombi in vessels of the lungs, kidneys and large bowel. The large bowel epithelium exhibited focal erosions; there was dense lymphocytic infiltration of the colonic lamina propria and submucosa, with a few plasma cells and eosinophilic leukocytes. Several reactive lymphoid follicles were present, with areas of polypoid hyperplasia of the colonic mucosa. The muscle coat and serosa showed mild lymphocytic infiltration. The appendix showed marked lymphoid hyperplasia. Crypt abscesses, granulomas, fissures and parasites were absent, and there was no evidence of lymphoproliferative neoplasia.

The fetus was a female whose development was consistent with the length of gestation, 37 weeks. She was 47 cm long and weighed 3010 g. The fetus was covered with vernix caseosa and showed meconium staining. Features of amniotic fluid aspiration were present, along with bilateral adrenal hemorrhage. There was no evidence of sepsis or microthrombi formation in any of the histologic sections. Death of the fetus was attributed to intrauterine anoxia.

The placenta was circumvallate; histologic sections showed no evidence of omphalitis, chorioamnionitis, placental inflammation (acute intervillositis) or macro- or microabscesses.

The two blood cultures were prepared in supplemented peptone broth (Vacutainer, Becton–Dickinson, Rutherford, New Jersey); they displayed turbidity, a brownish sediment and hemolysis after overnight incubation at 35°C.

Gram-negative bacilli and coccobacilli were found in primary cultures and after subculture on chocolate agar. Preliminary identification of *P. multocida* was made with the API 20E system (Analytab Products, Plainview, New York) and conventional methods. Confirmatory identification was done at the Central Laboratory, Ontario Ministry of Health, Toronto. Dr. Jacmohan L. Bhasin, of the National Research Council, Ottawa, further identified the strain as serotype 3, one of 16 known serotypes. The organism was shown to be susceptible to penicillin, ampicillin, trimethoprim–sulfamethoxazole, tetracycline, cephalothin, cefoxitin, cefotaxime, gentamicin, tobramycin and chloramphenicol in susceptibility tests with an agar dilution technique.

Table I — Selected characteristics of four cases of *Pasteurella multocida* septicemia during pregnancy

Characteristic	Robinson[1]	Kam et al[2]	Strand et al[3]	Present case
History of animal bites/scratches	None (but rats in vicinity)	Cat scratch	Cat scratch	Cat bite
Tissue reaction to bite/scratch	?	?	?	–
Gravida 1	?	+	–	+
Gravida 2	?	–	+	–
Third trimester	+	+	+	+
Abdominal pain	+	+	+	+
Nausea or diarrhea	?	+	?	+
Fever	+	+	+	+
Blood culture	+	+	+†	+
Septic arthritis	+	–	–	–
Premature rupture of membranes	+	–	+	–
Chorioamnionitis	?	–	+	–
Death of fetus	+	–	+	+
Fetus infected	?	–	?	–
Death of woman	–	–	–	+
Lymphoid bowel hyperplasia	?	+*	?	+‡
Several illness-related admissions	+	+	–	+

*Appendix only.
†Cord blood only.
‡Appendix and large bowel.

Discussion

Comparison of the clinical features of our case with those of three similar cases from the English-language literature[1-3] showed several common features: a history of animal bites or scratches, infection in the third trimester, fever, abdominal pain, several illness-related office visits or hospital admissions, positive results of blood cultures and death of the fetus (Table I).

We can only speculate as to the possible mechanisms of the fulminating illness in our patient. Humoral immunity apparently remains intact during pregnancy, with no significant changes in the levels of IgA, IgM and IgG.[4,5] In contrast, published studies indicate that the phagocytic and cell-mediated immune systems, two key elements in host resistance to bacterial infection, are attenuated.[6,7]

Pregnancy exerts an array of biologic effects on neutrophils. The absolute neutrophil count is increased during all stages of pregnancy,[8] and the cells are larger than normal,[9] but there is no information on their deformability. Random migration does not differ significantly between cells of pregnant women and normal cells in the presence of serum from pregnant women,[10] but neutrophil function is impaired by factors in such serum. Neutrophils from pregnant women exhibit a decreased chemotactic response to C5a (the activated fragment of complement component C5) and FMLP (*N*-formylmethionylleukophenylalanine).[11] While there is disagreement,[12] this effect also appears to occur with normal neutrophils from people exposed in vitro to serum from pregnant women;[13] however, the mechanism of this abnormal chemotactic response is at present unclear. Phagocytosis of *Escherichia coli*[10] and *Staphylococcus aureus*[14] also appears to be inhibited in pregnant women. This effect has been attributed to a heat-stable, nondialysable suppressor factor in their serum.[14] The phagocytic function of neutrophils from healthy males and pregnant women is altered to the same extent in the presence of serum from women in the third trimester of pregnancy.[14] In addition, intracellular killing of ingested bacteria is inhibited by serum from pregnant women,[14] but the mechanism is unknown. There is also an oxidative defect in neutrophils of pregnant women, as evidenced by a decrease in the production of superoxide anion in response to FMLP, but serum from pregnant women does not induce this effect in neutrophils from other people.[11]

Neutrophils are also an important target for bacterial endotoxins, which bind to neutrophils in vivo and in vitro. Endotoxins impair neutrophil adhesion, suppress locomotion and induce selective degranulation.[15] Oxidative and microbicidal properties may also be altered by complement-dependent and complement-independent mechanisms.[15]

There is convincing evidence that depressed cell-mediated immunity during pregnancy is responsible for susceptibility to new infections, reactivation of latent infections, increased clinical severity of certain infections and even a fatal outcome. Gestation-associated serum factors also cause alterations in lymphocyte and macrophage function. In this context, profound decreases in the ratio of helper to suppressor T cells have been reported in the third trimester, with no change in the absolute number of B cells.[6]

P. multocida infection should be suspected in all symptomatic patients with a history of animal bites or scratches. Blood and wound cultures are essential for definitive diagnosis. The absence of clinical signs, such as pain, erythema, swelling, dysfunction and regional lymphadenopathy, suggests impaired localization of infection; these signs may be absent in the pregnant patient during the third trimester. In our case the fetus showed no histopathologic evidence of infection, despite the fulminating, severe infection in the mother. Prompt antimicrobial treatment may be indicated for third-trimester patients with a history of animal bite.

The true incidence of animal bites that result in *P. multocida* infections in Canada is unknown, primarily because there is no statutory requirement for reporting bites that are infected, as determined bacteriologically. While reporting of infected bites is cumbersome, the US Air Force has established a precedent in the reporting and investigation of all bites.[16] This system, which includes coded ratings for severity of trauma, could be selectively used to identify virulent *P. multocida* infections in susceptible people.

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Reduction of prothrombin and partial thromboplastin times with trazodone

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Trazodone hydrochloride, a derivative of triazolpyridine, is an atypical, nontricyclic antidepressant.[1-3] Tricyclic antidepressants have long been known to interact with coumarin anticoagulants, increasing coagulation times.[4,5] We report a case of interaction between trazodone and warfarin in which trazodone reduced, rather than increased, the prothrombin and partial thromboplastin times.

Case report

A 40-year-old woman, who had been receiving 6.4 mg of warfarin daily after undergoing a mitral valve replacement, was treated with trazodone for severe depression. Her prothrombin and partial thromboplastin times had been 20 and 44 seconds respectively, but after 5 weeks of therapy with trazodone, the dose being gradually increased to 300 mg daily, they decreased to 14 and 36 seconds, decreases of 30% and 18% respectively. The daily dose of warfarin was increased to 7.5 mg, and the prothrombin and partial thromboplastin times returned to their original values.

Because of intolerable sedation and unsatisfactory therapeutic results, trazodone was replaced with doxepin hydrochloride, a tricyclic antidepressant. The daily warfarin dose was reduced to its original amount and 1 month later to 5.5 mg.

Comments

Trazodone is a weak selective inhibitor of serotonin reuptake,[1] but its inhibitory action is greater than that of amitriptyline hydrochloride.[1] It also has α-adrenergic blocking activity.[6]

Trazodone was thought to be less cardiotoxic than tricyclic antidepressants,[1,7] having no inotropic side effects,[1] and was consequently suggested as appropriate for treating patients with cardiac disease.[7] However, reports of trazodone-induced arrhythmias in patients without personal or family histories of cardiac disease have recently been published.[8,9]

Patients with cardiac problems frequently have associated depression,[6] and many are also receiving anticoagulant agents. Because of the trazodone–warfarin interaction, whose mechanism is still undefined, we recommend that the prothrombin and partial thromboplastin times be monitored three times weekly during the introduction of trazodone therapy, until the effects of the drug are optimal.

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